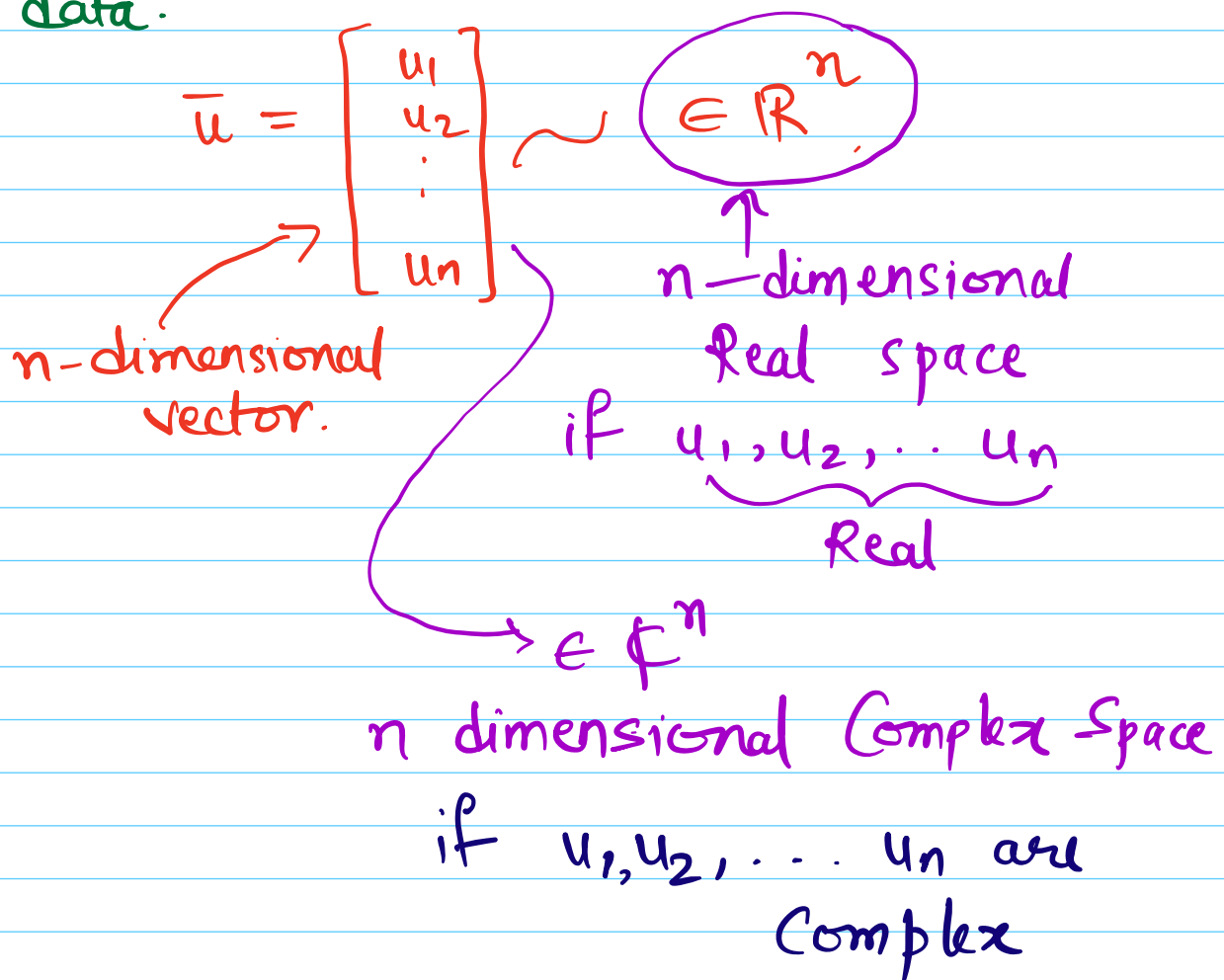


Lec - 01

Vector Properties : Addition, Linear combination,
Inner product, Orthogonality, Norm.

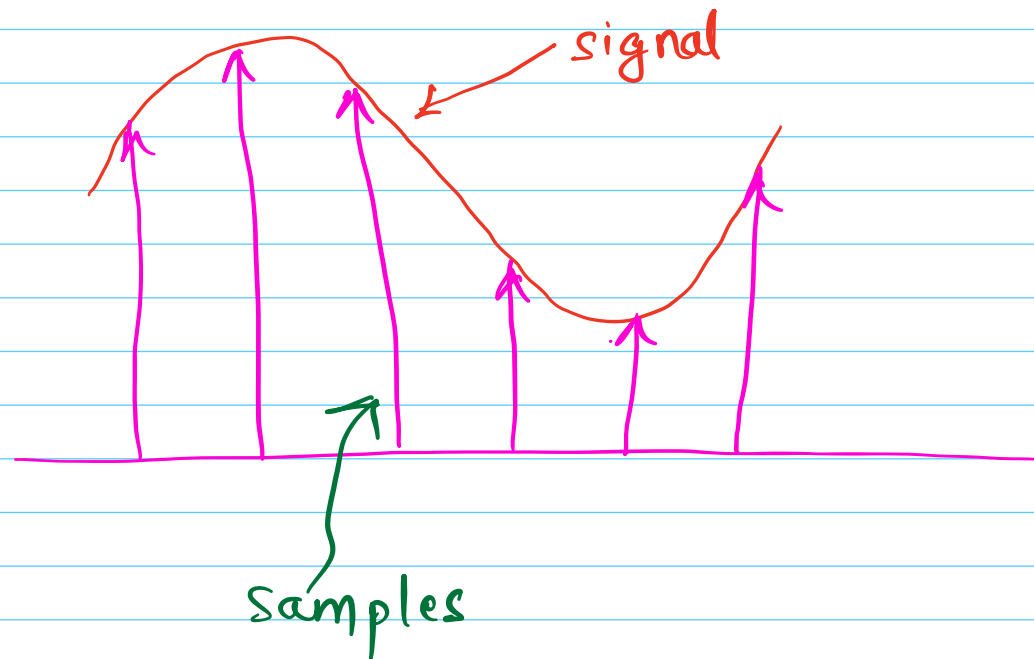
Signal Processing
Machine Learning
Data Analytics.

Vector is Fundamental to represent
data.



$\bar{u}$ represents an n-tuple
point in n-dimensional space

→ By default we will consider real vectors.



n samples of a signal.

n observations of a sensor
such as temperature,
pressure etc.

Location in 3D space

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

3-dimensional vector

Vector addition:

Consider 2 n -dimensional
vectors.

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$\vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

Scalar product:

$$\begin{array}{c} \nearrow \\ \text{scalar} \end{array} k \cdot \vec{u} = \begin{bmatrix} k \cdot u_1 \\ k \cdot u_2 \\ \vdots \\ k \cdot u_n \end{bmatrix}$$

Linear Combination of vectors

$\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ vectors

$k_1, k_2, \dots, k_n$ scalars

$$k_1 \vec{u}_1 + k_2 \vec{u}_2 + \dots + k_n \vec{u}_n$$

Linear combination.

very important concept

Inner Product

Consider arbitrary vectors

$$\bar{u}, \bar{v} \in \mathbb{R}^n.$$

$$\bar{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

$$\bar{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$\bar{u}^T = [u_1 \ u_2 \ \dots \ u_n]$$

→ Row vector.

$$\underbrace{\langle \bar{u}, \bar{v} \rangle}_{\substack{\text{inner} \\ \text{product}}} = \bar{u}^T \bar{v} = [u_1 \ u_2 \ \dots \ u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$= \sum_{i=1}^n u_i v_i$$

$$= u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

$$\text{Ex: } \bar{u} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \quad \bar{v} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

$$\begin{aligned}
 \langle \bar{u}, \bar{v} \rangle &= \bar{u}^T \bar{v} \\
 &= [2 \ 1 \ -1] \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} \\
 &= 2 - 1 - 3 = -2
 \end{aligned}$$

For complex vectors:

$$\begin{aligned}
 \langle \bar{u}, \bar{v} \rangle &= \bar{u}^H \cdot \bar{v} \\
 &= [u_1^* \ u_2^* \ \dots \ u_n^*] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\
 &= \sum_{i=1}^n u_i^* v_i \\
 &= u_1^* v_1 + u_2^* v_2 + \dots + u_n^* v_n
 \end{aligned}$$

ORTHOGONALITY.

→ Two vectors are orthogonal if their inner product = 0.

$$\mathbb{R}^n: \quad \bar{u}^T \bar{v} = 0.$$

$$\mathbb{C}^n: \quad \bar{u}^H \bar{v} = 0.$$

$$\text{Ex: } \bar{u} = \begin{bmatrix} 1 \\ -1 \\ +1 \\ -1 \end{bmatrix} \quad \bar{v} = \begin{bmatrix} 2 \\ 2 \\ -2 \\ -2 \end{bmatrix}$$

$$\underbrace{\bar{u}^T \bar{v}} = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ -2 \\ -2 \end{bmatrix}$$

$$= 2 - 2 - 2 + 2 = 0.$$

→ These 2 vectors are
ORTHOGONAL.

Complex sinusoids

$$\bar{u} = \begin{bmatrix} 1 \\ e^{j2\pi/N} \\ e^{j4\pi/N} \\ \vdots \\ e^{j2\pi(N-1)/N} \end{bmatrix} \quad \bar{v} = \begin{bmatrix} 1 \\ e^{j4\pi/N} \\ e^{j8\pi/N} \\ \vdots \\ e^{j4\pi(N-1)/N} \end{bmatrix}$$

Complex sinusoids

$$f_1 = \frac{1}{N}$$

$$f_2 = \frac{2}{N}$$

$$e^{j2\pi f_1 n}$$

$$e^{j2\pi f_2 n}$$

$$\bar{u}^H \bar{v} = \sum_{l=0}^{N-1} e^{-j2\pi l/N} e^{+j4\pi l/N}$$

$$= \sum_{l=0}^{N-1} e^{j2\pi l/N}$$

$$= \frac{1 - e^{j2\pi N/N}}{1 - e^{j2\pi/N}}$$

$$= \frac{1 - 1}{1 - e^{j2\pi/N}} = 0.$$

Complex sinuoids at Frequencies

$$0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}$$

→ orthogonal

→ This property is very important in Fourier Analysis.

Norm of a vector

↳ denoted by $\| \bar{u} \|$

$$= \sqrt{\langle \bar{u}, \bar{u} \rangle}$$

$$\begin{aligned} \langle \bar{u}, \bar{u} \rangle &= u_1^2 + u_2^2 + \dots + u_n^2 \\ &= \bar{u}^T \bar{u} \end{aligned}$$

$$\rightarrow \|\bar{u}\|_2 = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$$

Real vectors

length of a vector

$$\|\bar{u}\| = \sqrt{|u_1|^2 + |u_2|^2 + \dots + |u_n|^2}$$

Complex vector

Easy to see, $\|\bar{u}\| \geq 0$.

$$\hookrightarrow \|\bar{u}\| = 0 \Leftrightarrow \bar{u} = 0.$$

